

- 1) Experience has shown that 20% of a manufactured product is of top quality. In one day's production of 400 articles, Only 50 are of top quality. show that either the production of the day was chosen was not a representative sample or the hypothesis of 20% was wrong. Based on the particular day's production find also the 95% confidence limits of the percentage of the top quality product.
- 2) The family rate of typhoid is believed to be 17.26% . In a certain year 640 patients suffering from typhoid were treated in a metropolitan hospital and only 63 patients died .consider canyou the hospital efficient ?
- 3) A salesman in a departmental store daims that at most 60 percent of the shoppers entering the store without marking a purchase . A random sample of 50 shoppers showed that 35 of them without making a purchase. Are these sample results consistent with The claim of the salesman? Use an LOS of 0.05.
- 4) A cubical die is thrown 9000 times and a throw of 3 or 4 is observed 3240 times. show That the die cannot be regarded as an unbiased one. and find the extreme limits between which the probability of a throw of 3 or 4 lies.
- 5) A coin is thrown 400 times and is found to result in ' Head' 245 times Test whether the coin is a fair one.
- 6) A manufacturer of light bulbs claims that on the average 2% of the bulbs manufactured by his firm are defective. A random sample of 400 bulbs contained 13 defective bulbs On the basis of this sample can you support the manufacturer's claim at 5% LOS?
- 7) In a large city 20% of a roundom sample of 900 school boys had a slight defect. In another large city B 18.5% ofa random sample of 1600 boys had same defect. Is the difference between the proportions significant?
- 8) Before an excise duty on tea 800 people out ofa sample of 1000 were consumers of tea After the increase in duty 800 people were consumers of tea in a sample of 1200 persons. find whether there is significant decrease in the consumption of tea after the increase in duty.
- 9) 15.5% of a random sample of 1600 undergraduates were smokers whereas 20% random sample of 900 postgraduates were smokers in a state. can we conclude that less number of undergraduates are Smokers than postgraduates?
- 10) A sample of 300 spare parts produced by a machine contained in 48 defectives of Another sample 100 spare parts produced by another machine contained 24 defectives can you conclude that the first machine is better than the second?
- 11) A cigarette manufacturing company claims that its brand A cigarettes outsells its brand B by 8%. It is found that 42 out of a sample of 200 smokers prefer brand A and 18 out of another sample of 100 smokers prefer brand B. Test at 5% LOS whether the 8% valid difference is a claim
- 12) A sample of 100 students is taken from a large population students The mean height ofthe in this sample is 160 cm. can it be reasonably regarded that, in a population, 165m. the mean height is 165 Cm and the SD is 10 cm?
- 13) The mean breaking strength of the cables supplied by a manufacturer is 1800 with a SD of 100. By a new technique in the manufacturing process, it is claimed that the breaking cable strength of the has increased To test this claim a sample of 50 cables is tested and it is found that the mean strength breaking is 1850 .can we support the claim at 1% Los?
- 14) The rean value of a random sample of 60 items was found to be 145, with a SD of 40. Find the 95% confidence limits for the population mean. What size of the sample is required So estimate the population mean within 5 of its actual value with 95% or more confidence , using the sample mean ?
- 15) A sample of 400 observations has tmean 95 and SD 12. Could it be a random sample from a population with mean 98? What should be the massimeem Value of the population mean so that the sample can be regarded as one drown from it almost certainly?
- 16) A manufacturer claims that the mean breaking Strength of safety belts for air passengers produced in his factory is 1275 Kgs A sample of 100 belts were tested and the mean breaking strength and SD were found tobe 1258 and 90 kgs respectively Test the man ufactusers claim at 5%.Los.

17) In a random sample of size 500, the mean is found to be 20. In another independent sample of size 400, the mean is 15. could the sample been drawn from the same population with SD 4?

18) A simple sample of heights of 6400 English men has a mean of 170 cm and a SD of 6.4 cm, while a simple sample of heights of 1600 Americans has a mean of 172 cm and a SD of 6.3 cm. Do the data indicate that Americans are, on average, taller than the Englishmen?

19) Test the significance of the difference between the means of the samples, drawn from the two normal populations with the same SD Using the following data:

	size	Mean	SD
Sample 1	100	61	4
Sample 2.	200	63	6

20) The average marks scored by 32 boys is 12 with a SD of 8, while that for 36 girls is 70 with a SD of 6. Test at 1% LOS whether the boys perform better than girls.

21) The means of two simple samples of 1000 and 2000 items are 170 cm. and 169 cm respectively.

• can the samples be regarded as drawn from the same population with SD 10 at 5% Los?

22) In a college 60 junior students are found to have a mean height of 171.5 cm and 50 senior students are found to have a mean height of 173.8 cm. Can we conclude that the Juniors based on these data, are shorter than seniors at (i) 5% LOS and (ii) 1% Los, assuming that the SD of students of that college is 6.2 cm?